

Eastern Mediterranean University – Computer Engineering Department

CMPE/CMSE/AING231 – Data Structures

Assignment #3

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Cyprus bank needs a queue management program. When queuing incoming customers, there should be a program that will take the names of the customers as input and display the name of the customer whose turn it is. It should also provide a list of pending customers when prompted.

You should write a menu-driven C program. Menu items can be:

- **Customer entrance:** The customer's name will be entered and put into the queue.
- **Customer exit:** The next customer will be removed from the queue and her/his name will be displayed.
- **Pending customers:** The names of all pending customers will be shown in the order of waiting.
- **Exit:** The program will be terminated.

Here are the points that you need to pay attention to:

- Your solution should be with a **queue data structure**.
- Many customers can enter, many customers can exit, and these entrances and exits can also be in a mixed order.
- When the queue is empty and the customer is tried to exit, the user should be warned with an appropriate message, but the program should **NEVER** be exited.
- When the queue is full and the customer is tried to enter, the user should be warned with an appropriate message, but the program should **NEVER** be exited. **The queue can be 10 maximum.**
- The only exit from the program should be by selecting the **"Exit"** option.
- Listing the pending customers should be done **by removing from the queue**; however, when the listing is done, the queue should **remain the same!**

Hint: Use a separate function for listing, and the function parameter should be queue structure but not a **pointer** to queue structure. Therefore, any modification you make within the function will not affect your original queue content.

NOTE: If any of the following situations occur, you will get ZERO:

- You have not uploaded your solution by the deadline.
- You have produced a solution other than C language.
- You have uploaded an incorrect file or a file that cannot be opening.
- You have not used queue data structure as mentioned above.
- Your solution contains **even a single line of code** that is NOT yours.
- **You have shared your solution or a part of your solution with someone else!!**